

# MRI-to-CT Cross-Modality Image Synthesis using GANs

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## Members:

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## Introduction:

In medical imaging, multiple modalities are often required to provide complementary diagnostic information. **Magnetic Resonance Imaging (MRI)** excels in visualizing soft tissues, while **Computed Tomography (CT)** provides superior imaging of bone and is widely used for treatment planning. However, acquiring both scans can be time-consuming, costly, and exposes patients to ionizing radiation.

## Problem Statement:

This project aims to develop a **Generative Adversarial Network (GAN)-based model** that synthesizes CT images directly from MRI scans. By doing so, the need for an additional CT scan can be reduced, thereby minimizing patient exposure to radiation, lowering costs, and saving time.

## Reason for Selection:

This problem is well-defined, clinically meaningful, and feasible within the timeframe of the course. Public datasets are available (e.g., CHAOS Challenge dataset), and the project leverages well-established deep learning architectures such as **Pix2Pix** and **CycleGAN**.

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## Methods:

### Approach:

- Train and evaluate a **GAN-based model** to perform MRI-to-CT cross-modality image synthesis.
- Explore both **Pix2Pix** (paired data) and **CycleGAN** (unpaired data) architectures:
  - **Pix2Pix**: Requires paired MRI and CT images of the same patient.
  - **CycleGAN**: Allows training on unpaired datasets, which are easier to acquire.

## Research Questions:

1. Can GAN-based models generate CT-like images from MRI scans with high anatomical fidelity?
2. Does CycleGAN, which uses unpaired datasets, perform comparably to Pix2Pix trained on paired data?

## Software Platforms & Tools:

- Programming Language: **Python**
- Deep Learning Framework: **PyTorch**
- Libraries: Numpy, Scikit-image, OpenCV, Matplotlib
- Training Monitoring: TensorBoard

- **GUI Requirement:** A simple Python/MATLAB interface to input MRI images and view synthetic CT outputs.

#### **Dataset Sources:**

- **CHAOS Challenge Dataset:** Provides multi-modality abdominal scans (MRI + CT), suitable for paired approaches.
- If paired datasets are limited, we will train using **CycleGAN** with unpaired MRI (e.g., ADNI dataset) and CT datasets (e.g., LIDC-IDRI).

#### **Evaluation Metrics:**

- **Quantitative:** Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM).
- **Qualitative:** Visual comparison of anatomical detail preservation and artifact generation.

#### **Presentation of Results:**

- Comparative visualization of input MRI, synthetic CT, and ground truth CT.
- Tables summarizing PSNR/SSIM across test samples.
- GUI demo of MRI-to-CT synthesis.

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#### **Challenges:**

- **Data Limitations:** Paired MRI-CT datasets are relatively scarce. CycleGAN helps mitigate this by using unpaired data.
- **GAN Training Stability:** Training adversarial networks can be unstable and requires careful hyperparameter tuning.
- **Evaluation of Realism:** Numerical metrics may not fully capture diagnostic usability; qualitative inspection is important.
- **Computational Resources:** GAN models require GPUs and sufficient training time.

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#### **Literature Review:**

1. Zhu et al., “*Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks (CycleGAN)*.” arXiv:1611.07004.
2. Isola et al., “*Image-to-Image Translation with Conditional Adversarial Networks (Pix2Pix)*.” arXiv:1611.07004.